

# Talos

An open source multi agent system for automated attack classification and scope finding

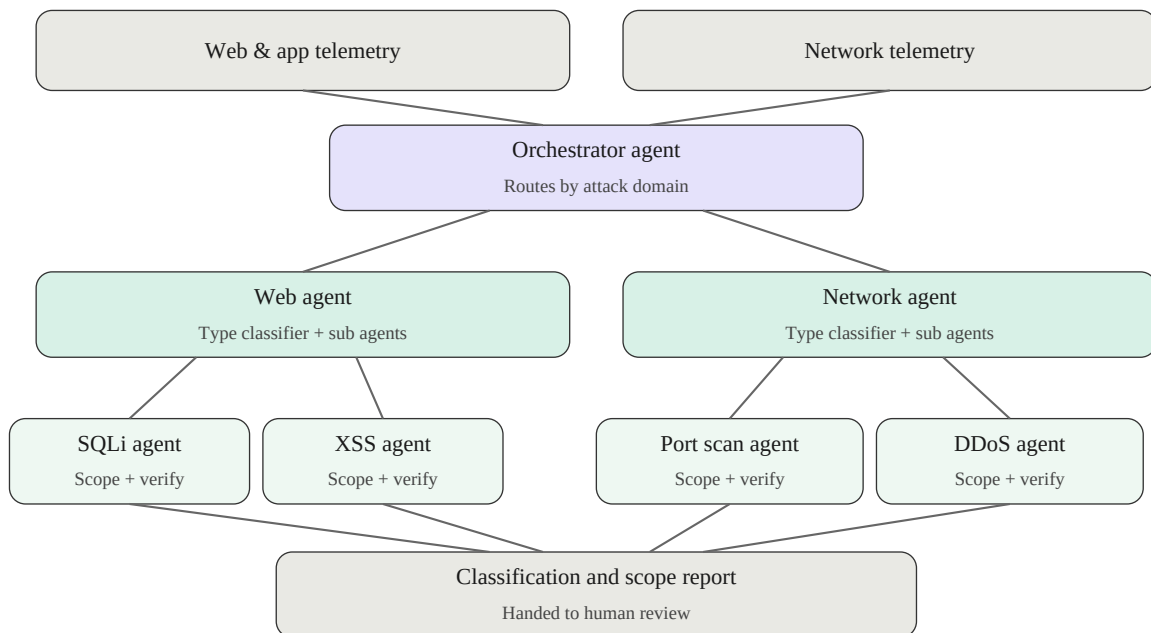
## The problem

Security teams are overwhelmed by alerts from web application and network sources. Manually detecting, classifying, and scoping each potential attack consumes most of the time in incident response. Existing AI powered SOC platforms are largely closed source, enterprise priced, and by their own vendors' admission strong only at surface level first stage triage, with deeper contextual investigation remaining weak across the industry.

## The idea

Talos is an orchestrator driven multi agent system. An orchestrator agent routes incoming telemetry to the right domain. Within each domain, a lightweight classifier identifies the likely attack type, and only then activates the relevant specialized sub agent, which confirms the classification and determines the exact scope of the attack, such as which endpoints, accounts, or data were affected. Every reasoning step is visible and inspectable, unlike a single opaque model producing a verdict.

## Architecture



*Note: the diagram above shows a simplified sample of the sub agent set for illustration. The complete, detailed architecture diagram covering the full sub agent set is provided as a separate attachment.*

## Scope for this build

Full working implementation across two domains, web and network, with real detection logic grounded in the OWASP Top 10 framework. Initial focus categories include Injection (SQL injection and cross site scripting), Authentication Failures, and Broken Access Control for the web domain, and port scanning, DDoS, and brute force detection for the network domain. Additional categories are added incrementally as an extensible, community contributable framework, validated against known vulnerable open source applications such as OWASP Juice Shop.

## Why it is different

**Open and transparent**

Every classification and scoping decision is inspectable, not a black box verdict

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| <b>Depth over breadth</b>      | Real working detection logic for a focused, OWASP aligned set of attack types        |
| <b>Domain agnostic core</b>    | Same orchestrator and classify then dispatch pattern extends across domains          |
| <b>Built for smaller teams</b> | A free, extensible alternative for teams that cannot afford enterprise SOC platforms |

## Technology stack

Python for core agent logic, LangGraph and LangChain for multi agent orchestration, a Claude or GPT class model for the reasoning layer of each agent, FastAPI for service exposure, PostgreSQL for storing alerts and reports, and Docker for packaging. OWASP Juice Shop and DVWA are used to generate and validate real attack traffic during development.